

2 ways to create an array

1) without using new keyword

2) with using new keyword

1-D array , 2-D arrays

2-D array creation with new keyword

```
int marks[][] = new int[3][5];
```

```
marks[0][0] = 50;
```

```
marks[0][1] = 60;
```

```
marks[0][2] = 78;
```

```
marks[0][3] = 87;
```

```
marks[0][4] = 95;
```

```
System.out.println(Arrays.deepToString(marks));
```

Assigning one Array to Another Array

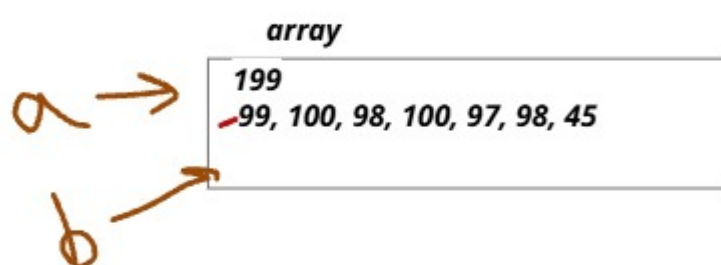
```
int a[]
    = { 99, 100, 98, 100, 97, 98, 45 };

int b[] = a;

System.out.println(Arrays.toString(a));
System.out.println(Arrays.toString(b));

a[0] = 199;

System.out.println(Arrays.toString(a));
System.out.println(Arrays.toString(b));
```



When we assign one array in java to another array, then single array itself gets created and java will make both the new array names point to the same array

How to copy elements from one array to another array?

```
int marks[] = new int[3];
```

```
marks[0] = 34;
```

```
marks[1] = 44;
```

```
marks[2] = 54;
```

```
int finalMarks[] = Arrays.copyOf(marks, 5);
```

```
System.out.println(Arrays.toString(marks));
```

```
System.out.println(Arrays.toString(finalMarks));
```

Vector**Arrays and Vector ==> 99% same****1) Write a program to store the marks of 10 students**

```
int marks = new int[10];
```

2) Write a program to store the marks of all students in the class

```
int marks = new int[153];
```

Vector ==> It is similar to arrays, while creating the array we should specify the size of the array, but while creating vector we need not specify any size
vector comes with a default size of 10

when to use arrays and when to use vector?

1) we use arrays when we know how many elements we have to store

1) we use vector when i want to store some elements but i dont know the count

2) we use arrays when we want to store same type of data

2) we use vector when we want to store same type of data OR different type of data

NOTE: we can make vector to store only certain type of data, we use GENERICS(<Integer> OR <String>,)